

NICOLETTE MASHABA

Junior AI Software Engineer · Azure Durable Functions · RAG · DSPy · LLMs · AZ-204
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[LinkedIn](#) | [GitHub](#) | [Portfolio](#)

PROFESSIONAL SUMMARY

Junior AI Software Engineer with hands-on production experience building AI agent pipelines, RAG systems, and compliance automation on Microsoft Azure. Currently engineering multi-agent architectures using DSPy, LangChain, and Azure Durable Functions at Always Enough LLC, the technology company behind Marisa Peer's RTT brand. AZ-204 certified with a proven track record delivering real-world AI solutions — including a hackathon-winning compliance tool featured on an international technology podcast. Skilled in prompt engineering, secure development practices, GDPR-aware data handling, and translating complex business problems into scalable, maintainable software.

TECHNICAL SKILLS

Languages: Python, JavaScript, TypeScript, C#, SQL

Frontend: React, HTML5, CSS3, Tailwind CSS, Responsive Design

Backend: Node.js (Express), FastAPI, ASP.NET Core, REST API Design

AI & Automation: OpenAI API, GPT-4, LLM Integration, Prompt Engineering, RAG Pipelines, DSPy, LangChain, LangGraph, Azure Durable Functions, ChromaDB, ElevenLabs API, Structured Outputs

Databases: MongoDB Atlas, SQL Server, PostgreSQL, Firebase, ChromaDB (vector database)

Cloud & Deployment: Microsoft Azure (AZ-204), Azure Functions, Azure Blob Storage, Azure Communication Services, Azure DevOps, Vercel, Render, Netlify, Docker (fundamentals)

Testing & QA: Manual Testing, Bug Triage, API Testing (Postman)

Tools & Practices: Git, GitHub, CI/CD Pipelines, Azure DevOps, Agile/Scrum, Environment Variables, GDPR Awareness

In Progress: Azure DevOps Engineer Expert (AZ-400) — actively building knowledge

PROFESSIONAL EXPERIENCE

Junior AI Software Engineer

Apr 2026 – Present

[Always Enough LLC](#) · Dubai, UAE · Remote

- Designed and implemented Azure Durable Function timer triggers replacing a Next.js heartbeat scheduler, improving pipeline reliability for a production sales call analysis system.
- Built and deployed a compliance analysis pipeline using GPT-4 and Azure AI to automatically score sales calls against regulatory guidelines, delivering structured JSON reports with violation flagging and mandatory statement verification.
- Diagnosed and resolved a UniqueViolation production bug in PostgreSQL by replacing plain INSERT logic with upsert operations, eliminating duplicate record crashes in the transcription pipeline.
- Implemented RAG architecture using ChromaDB and OpenAI embeddings for a personalised hypnosis audio generation system, significantly reducing script generation time.
- Researched and prototyped DSPy integration with LangChain for multi-agent copywriting systems, delivering working modules for landing page and email sequence generation.
- Worked within Azure DevOps using Git branching, pull requests, and CI/CD pipelines to deploy production code across Azure Functions, Blob Storage, and PostgreSQL on Azure.

Technical Trainer (Volunteer)

2026 – Present

TechBridle Foundation · South Africa · Remote

- Mentor learners in web development, JavaScript, React, and backend engineering concepts.
- Lead code review sessions and guide Git workflows, secure project setup, and debugging practices for junior developers.

Software Engineering Intern

2025

Coast IT (Pty) Ltd · South Africa

- Contributed to feature development within an Agile/Scrum team, participating in sprint planning, daily stand-ups, and retrospectives.
- Performed API testing using Postman, resolved bugs, and participated in peer code reviews to improve application stability and code quality.
- Collaborated with senior engineers to implement and test new application modules, gaining exposure to production-level codebases.

AI Model Trainer

2024

Outlier AI · Remote

- Evaluated AI-generated outputs and improved response quality through structured, documented feedback aligned with model guidelines.
- Conducted prompt testing and model behaviour analysis, identifying patterns that improved response clarity, accuracy, and reliability.

Junior Navigation Officer

2015 – 2021

South African Navy · South Africa

- Operated in mission-critical environments demanding precision decision-making, high ownership, and strict adherence to protocol — transferable qualities applied directly to software engineering.
- Developed cross-functional team coordination and communication skills under high-pressure, time-sensitive conditions.

PROJECTS

AIMS Compliance Interrogator *Hackathon Winner · Featured on International Tech Podcast*

- Identified inefficiencies in manual compliance review and architected an AI-powered tool to analyse structured inputs and automatically flag risks.
- Designed controlled LLM prompt workflows to ensure consistent, auditable, and explainable outputs — applicable to regulated AI environments.
- Enforced backend-only AI calls and secure environment configuration to handle sensitive compliance data safely.
- Deployed full-stack application on Microsoft Azure; won hackathon and was featured on an Australian technology podcast.

Stack: Python, FastAPI, OpenAI API, Microsoft Azure, Prompt Engineering

EmpowaAI *AI-Powered Career Platform · Startup · Active Development*

Co-building an AI-powered platform helping South African youth navigate career paths, optimise job readiness, and access economic opportunities through intelligent systems.

- Designed and developed the full-stack application integrating React (TypeScript), Node.js/Express, and FastAPI-based AI microservices, orchestrating reliable data flow across the entire stack.
- Engineered the Digital Economic Twin concept to model user profiles and generate personalised career insights and income pathway simulations.
- Built AI-powered features including CV analysis, CV revamp, interview coaching, and skill-based opportunity matching — deployed to beta users on Vercel and Render.
- Iterated on features based on real user feedback, improving AI response relevance through structured prompt engineering and A/B testing of prompt variants.
- Contributed to product architecture decisions in a startup environment, balancing delivery velocity with code quality.

Stack: *React, TypeScript, Node.js, Express, FastAPI, MongoDB Atlas, OpenAI API, Tailwind CSS, Vercel, Render*

BookReviewApp *Full-Stack Web Application*

- Built a full-stack book review platform with user authentication, content moderation, and full CRUD functionality.
- Designed RESTful APIs using ASP.NET Core and Entity Framework Core; implemented database migrations and clean architecture for scalability and data integrity.

Stack: *C#, ASP.NET Core, Entity Framework Core, SQL Server, REST APIs*

Interview Replay *Live Production Application*

- Developed an AI-driven mock interview tool delivering structured, actionable feedback on candidate strengths and improvement areas.
- Iteratively improved AI response quality through prompt engineering; deployed and maintained in production using clean Git practices and secure environment configuration.

Stack: *JavaScript, OpenAI API, Netlify*

CERTIFICATIONS & PROFESSIONAL DEVELOPMENT

- Microsoft Certified: Azure Developer Associate (AZ-204)
- Microsoft Certified: Azure Data Fundamentals (DP-900)
- J.P. Morgan Software Engineering Virtual Experience
- Azure DevOps Engineer Expert (AZ-400) — *knowledge in active development*

EDUCATION

Occupational Certificate: Software Engineering (NQF Level 6) CTU Training Solutions, South Africa	2026
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Information Technology: Programming Foundations (NQF Level 4) CTU Training Solutions, South Africa	2024
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